

In general, transactions per second, transaction latency and resource utilisation are the parameters to be examined for performance.

It is also extremely important to measure if the integrity of the network and the shared ledger is maintained while testing the throughput. There are no concrete complete tools available in the market, but HyperLedger Caliper and Jmeter can be used for performance tests to benchmark and conduct API tests, respectively.

Security validation: Vulnerability assessment should be done for all the nodes in the networks, be it a private or public blockchain network. Testing for the security on the network, and validating the privacy of the data in the distributed ledger should be ensured. Password strength, hardening of private keys, and private key storage implementation should be reviewed and tested.

There are a few challenges in blockchain testing, though.

- Testing blockchain applications requires multiple skills, including automation skills in all languages, understanding mid- and low-level integration, API and UI level testing skills, etc.
- Setting up the environment for the blockchain testing itself is a challenge.
- Getting the test and test automation right the first time is a real challenge, as updation and rollback on the contracts

is almost impossible. This is a huge challenge for the quality assurance team.

- Scenarios like tampering with the blocks as well as the size of the blocks pose a challenge. **END** 🐧

References

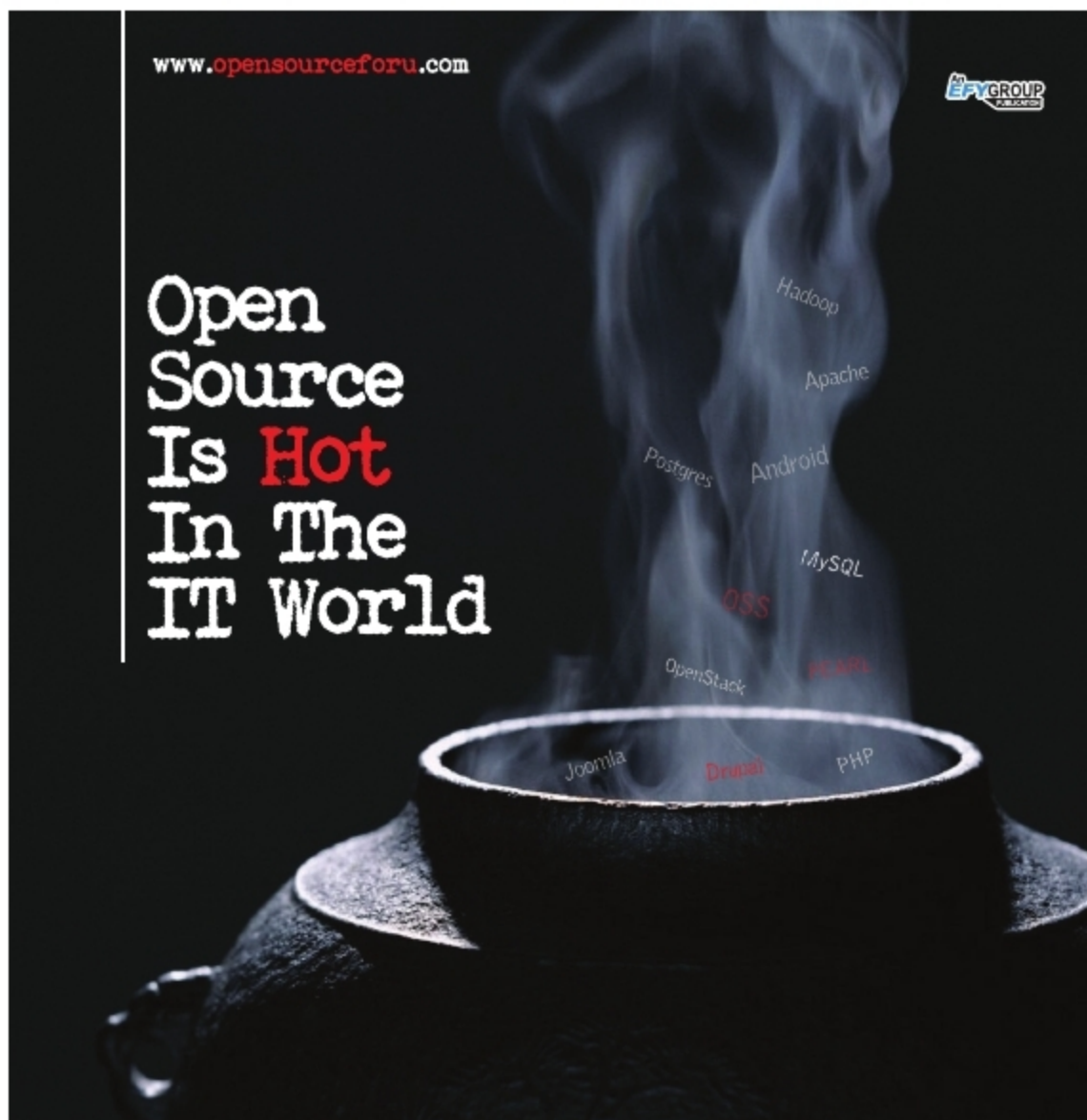
- [1] <https://www.joecolantonio.com/blockchain-testing-tools/>
- [2] <https://dzone.com/articles/test-driving-a-blockchain-application-a-simple-exa>
- [3] <https://www.infoq.com/articles/blockchain-middleware/>
- [4] <https://www.researchgate.net>
- [5] <https://www.coindesk.com/information/what-is-blockchain-technology>
- [6] <https://en.wikipedia.org/wiki/Blockchain>

By: Srikanth Mohan, Ganesan Nagasamy and Shashidhar Soppin

Srikanth Mohan is a senior architect with 15+ years of experience in IT. He specialises in test automation and performance testing.

Ganesan Nagasamy has 18+ years of experience in the software industry. He specialises in test automation, non-functional testing, the cloud, Docker, Agile, etc.

Shashidhar Soppin is a senior architect with 17+ years of experience in the IT industry. He specialises in virtualisation, Docker, the cloud, AI-ML, deep learning and OpenStack.



THE COMPLETE MAGAZINE ON OPEN SOURCE



- >> To find dealers to buy copy from news-stand, visit: www.ezine.lfyimag.com
- >> To subscribe, visit: pay.efyindia.com
- >> To buy an ezine edition, visit: www.magzter.com choose Open Source For You